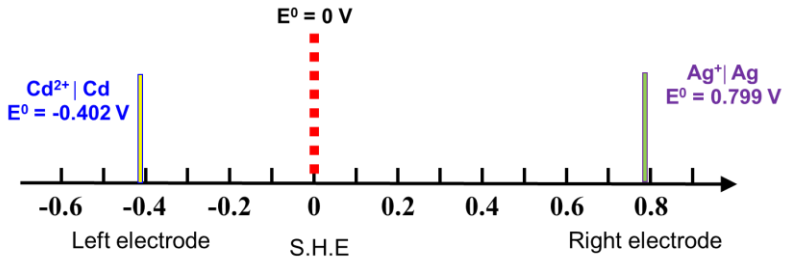
Analytical chemistry 1 for engineers

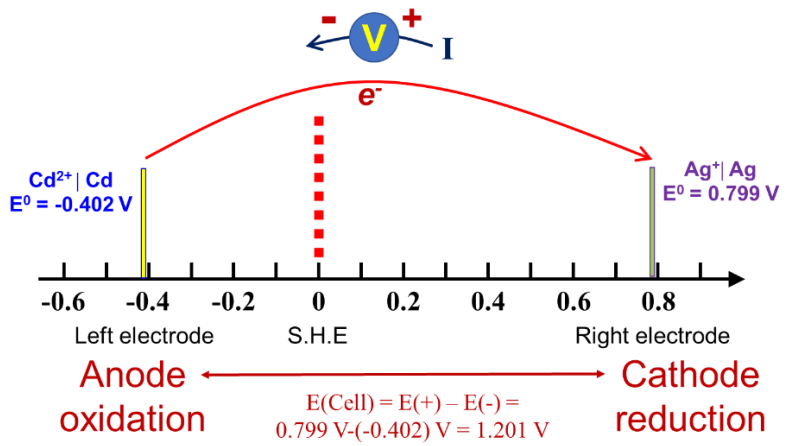
Homework 9 – Galvanic cell

姓名： **Name:** **ID number:** **Major:**

H9-1. Please mark the anode and cathode for the following galvanic cell. Write down the reactions (i.e. reduction or oxidation) on each electrode. Then, figure out the direction for the migration of electron (*e*) and current (I). Finally, calculate potential for the whole galvanic cell.



Solution:



Anode: in a galvanic cell, the anode is the electrode at which the oxidation reaction occurs and the electrons leave away.

Cathode: the electrode that the reduction reaction occurs and the electrons enter.

Usually, the anode is placed on the left side, and the cathode is placed on the right side.

Electrons flow through the circuit from the more negative to the positive electrode. However, the current always flows from the high potential to the low one.
The cell potential equals to the reduction potential ($Ag^+ | Ag$) on cathode substrates the reduction potential ($Cd^{2+} | Cd$) NOT ($Cd | Cd^{2+}$) on anode.

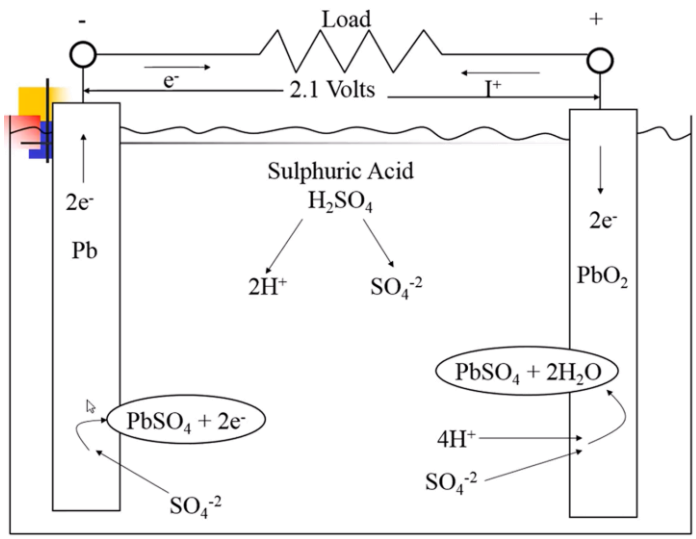
H9-2. The Nernst equation is used to obtain the potential of a half-reaction as a function of the concentrations of the chemical species taking part in. It is not possible to measure the absolute potential value of redox pair, but rather its difference relative to another, measured by reference half-cell (called a reference electrode)).
Which characteristic is about the reference electrode (Single choice question),

a. Universal acceptance
b. Compliance with Nernst's equation
c. Reproducibility
d. Hard to provide a constant potential

Answer: d
The reference electrode should be capable of providing a constant potential under certain conditions, e.g. saturated solution to make the value of log*x* be constant.

H9-3. Lead-acid battery was invented in 1859 by the French physicist Gaston Plante, which is the first rechargeable battery. Its electrodes are metallic lead grids with large surface area. Solid PbO_2 is pressed onto the cathode. The cell is filled with aqueous H_2SO_4, which is ~35 wt% (4.4 M) when fully charged. As the cell discharged, both electrodes become coated with $PbSO_4$ (*s*) as shown below. Both reactions consume H_2SO_4, whose concentration can fall to ~22 wt% (2.3 M) during discharge.

a. What's the reaction on the anode;
b. What's the reaction on the cathode;
c. Write down the net reaction.



Anode: $Pb(s) + SO_4^{2-} \rightleftharpoons PbSO_4(s) + 2e$

Cathode: $PbO_2(s) + 4H^+ + SO_4^{2-} + 2e \rightleftharpoons PbSO_4(s) + 2H_2O$

Net reaction: $Pb(s) + PbO_2(s) + 2H_2SO_4 \rightleftharpoons 2PbSO_4(s) + 2H_2O$

H9-4. Suppose that the concentration of NaF and KCl are 0.10 M in the following cell, respectively.

$$Pb(s) \mid PbF_2(s) \mid F^-(aq) \parallel Cl^-(aq) \mid AgCl(s) \mid Ag(s)$$

$$E^o(PbF_2 \mid Pb) = -0.350\ V$$

$$E^o(AgCl \mid Ag) = 0.222\ V$$

a. Write down the reactions on anode, cathode and total reaction, respectively;

b. Calculate the cell voltage;

c. In which direction do the electrons flow through the circuit?

a) Anode: $Pb + 2F^- \rightleftharpoons PbF_2(s) + 2e$

Cathode: $AgCl + e \rightleftharpoons Ag + Cl^-$

Total reaction: $Pb + 2F^- + 2AgCl \rightleftharpoons PbF_2(s) + 2Ag + 2Cl^-$

b) $E_{cell} = E^o(Pb \mid PbF_2) + E^o(AgCl \mid Ag) - \frac{0.05916}{2} \log\frac{[Cl^-]^2}{[F^-]^2}$

$= -E^o(PbF_2 \mid Pb) + E^o(AgCl \mid Ag) - \frac{0.05916}{2} \log\frac{[Cl^-]^2}{[F^-]^2}$

$= -(-0.350\ V) + 0.222\ V - \frac{0.05916}{2} \log\frac{(0.1\ M)^2}{(0.1\ M)^2}$

$= 0.572\ V$

c) Electrons flow from Pt electrode to Ag electrode.

H9-5. There is a galvanic cell in the line notation as following:

$$Pt(s) \mid Br_2(l) \mid HBr(aq, 0.10\ M) \parallel Al(NO_3)_3(aq, 0.01\ M) \mid Al(s)$$

$$E^o(Br_2 \mid Br^-) = 1.078\ V$$

$$E^o(Al^{3+} \mid Al) = -1.677\ V$$

a. Calculate the potential of each half cell and the cell potential;

b. In which direction will electrons flow through the circuit?

c. Write down the spontaneous net cell reaction;

d. The left half-cell was loaded with 14.3 mL of Br_2 (l) (density is 3.12 g/mL). The aluminum electrode contains 12.0 g Al. Which element, Br_2 or Al is the limiting reagent? (M.W. of Br is 79.904 g/mol, M.W. of Al is 26.981 g/mol)

a) For the left electrode:

$$Br_2(l) + 2e \rightleftharpoons 2Br^-$$

$$E_{left} = E^o - \frac{0.05916}{2}\log[Br^-]^2 = 1.078\ V - \frac{0.05916}{2}\log(0.10\ M)^2 = 1.137\ V$$

For the right electrode:

$$Al^{3+} + 3e \rightleftharpoons Al$$

$$E_{right} = E^o - \frac{0.05916}{3}\log\frac{1}{[Al^{3+}]} = -1.677\ V - \frac{0.05916}{3}\log\frac{1}{0.01\ M} = -1.716\ V$$

$$E_{cell} = E_{right} - E_{left} = -1.716\ V - 1.137\ V = -2.853\ V$$

b) Electrons flow from Al electrode to Pt electrode (from right to left). This is why the cell potential is minus. This notation is opposite to our default form.

c) Net cell reaction:

$$Al + \frac{3}{2}Br_2(l) \rightleftharpoons Al^{3+} + 3Br^-$$

d) The mole of $Br_2(l)$ is $\frac{14.3\ mL \times 3.12\ g/mL}{79.904\ g/mol} = 0.2792$ mol, the mole of Al is $\frac{12.0\ g}{26.981\ g/mol} = 0.4448$ mol

1 mol of Al requires 1.5 mol of $Br_2(l)$ in the net reaction. Therefore, 0.4448 mol of Al ask for 0.6672 mol of $Br_2(l)$ > 0.2792 mol.

So, $Br_2(l)$ is the limiting reagent in this system.